Chapter 3, Problem 93CP

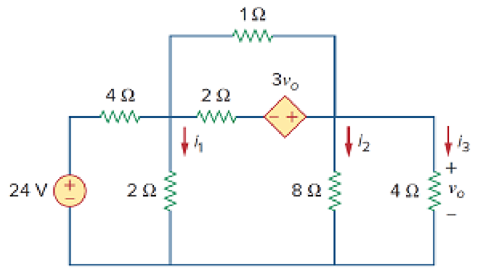
Problem

Rework Example 3.11 with hand calculation.

Example 3.11:

In the circuit of Fig. 3.34, determine the currents *i*1, *i*2, and *i*3.

Figure 3.34:

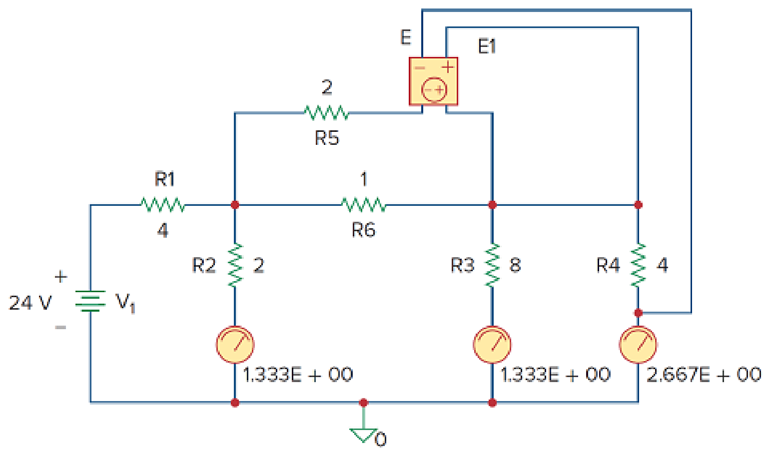


Solution:

The schematic is shown in Fig. 3.35. (The schematic in Fig. 3.35 includes the output results, implying that it is the schematic displayed on the screen *after* the simulation.) Notice that the voltage-controlled voltage source E1 in Fig. 3.35 is connected so that its input is the voltage across the 4-Ω resistor; its gain is set equal to 3. In order to display the required currents, we insert pseudocomponent IPROBES in the appropriate branches. The schematic is saved as *exam311.sch* and simulated by selecting **Analysis/Simulate**. The results are displayed on IPROBES as shown in Fig. 3.35 and saved in output file *exam311.out*. From the output file or the IPROBES, we obtain *i*1 = *i*2 = 1.333 A and *i*3 = 2.667 A.

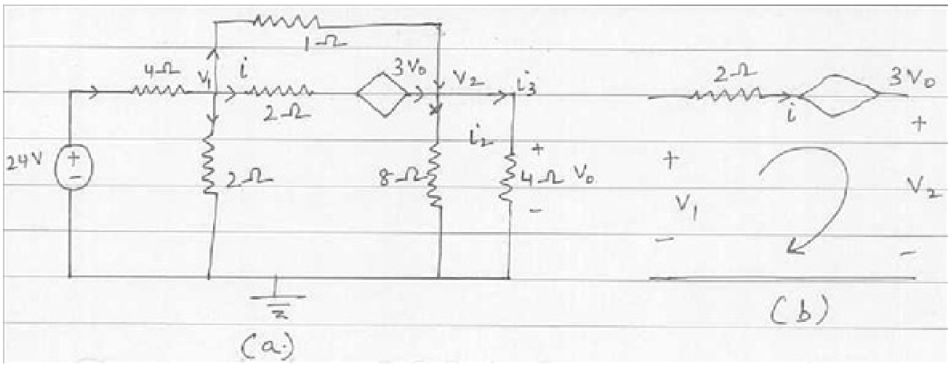
Figure 3.35

The schematic of the circuit in Fig. 3.34.



Step-by-step solution

Step 1 of 1



From (b),

$$-V_1 + 2i - 3V_0 + V_2 = 0 \quad \textit{Which leads to} \quad i = \frac{(V_1 + 3V_0 - V_2)}{2}$$

At node 1 in (a),

$$\left(\left(\frac{24 - V_1}{4}\right)\right) = \left(\frac{V_1}{2}\right) + \left(\frac{(V_1 + 3V_0 - V_2)}{2}\right) + \left(\frac{(V_1 - V_2)}{1}\right),$$

Where V0 = V2 Or,

24 = 9 V, Which leads to V1 = 2.667 Volts.

At node 2,

$$\left(\left(\frac{V_1 - V_2}{1}\right)\right) + \left(\frac{(V_1 + 3V_0 - V_2)}{2}\right) = \left(\frac{V_2}{8}\right) + \left(\frac{V_2}{4}\right), \quad V_0 = V_2,$$

$$V_2 = 4V_1 = 10.66\,Volts.$$

Now we can solve for the currents,

$$i_1 = \frac{V_1}{2} = 1.333\,A,$$

$$i_2 = 1.333\,A \quad \textit{And}$$

$$i_3 = 2.6667\,A.$$